

The same unmarked filtered complex

$$P_0 = H_0 = \mathbb{Q}^4, \quad P_i = 0 \ (i \neq 0)$$
$$\text{Descending flag: } \mathbb{Q}^4 \supset \langle e_2, e_3, e_4 \rangle \supset \langle e_3, e_4 \rangle \supset \langle e_4 \rangle \supset 0$$

Two filtration-preserving operators

$$Se_i = e_{i+1} \ (i < 4), \quad Se_4 = 0, \quad W = \text{diag}(0, 1, 2, 3)$$
$$S^4 = 0, \quad [W, S] = S$$

Choose the marking $\{S\}$

$$\mathbb{Q}[S] = \text{span}(I, S, S^2, S^3)$$
$$\dim \mathbb{Q}[S] = 4$$

Choose the marking $\{S, W\}$

The lower-triangular matrix algebra

$$\dim \mathbb{Q}\langle S, W \rangle = 10$$

Distinct weights recover the diagonal units; $E_{ii}S^{i-j}E_{jj} = E_{ij}$ for $i \geq j$.

Scope boundary

The bare filtered object does not determine which marking was intended.
Supplied marked chain maps do determine their effective algebra on H_0 .

This does not exclude intrinsically distinguished scalar maps such as I ,
or additional geometric data that select further operators.

Change of basis gives conjugate represented algebras. A contractible summand
has no effect when the descended marked actions are identified.